Week 3 · Part A — nine charts, rebuilt and taken apart

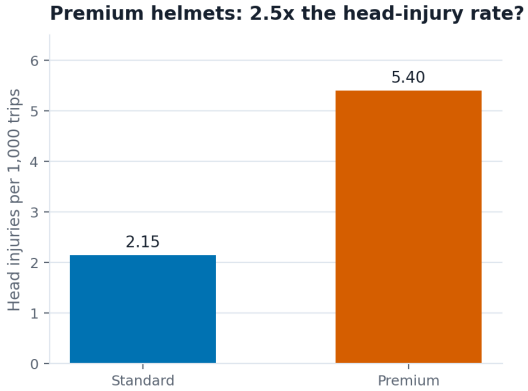
Every chart below was rebuilt from its recipe (*partA_charts.py*) and hit its self-check number. Each case gives the conclusion and decoy, the mechanism with the number that proves it, and the redesign. All data is synthetic, generated for CDB601220 Week 3.

Case 1 · Premium helmets and head injuries

“Premium-certified helmets have 2.5x the head-injury rate.”

1 · Rebuild

Bar chart, y from zero, one bar per grade (the recipe).



y-axis from zero; 100,000 trips. Synthetic data generated for CDB601220 Week 3.

2 · The conclusion the chart invites, and the decoy

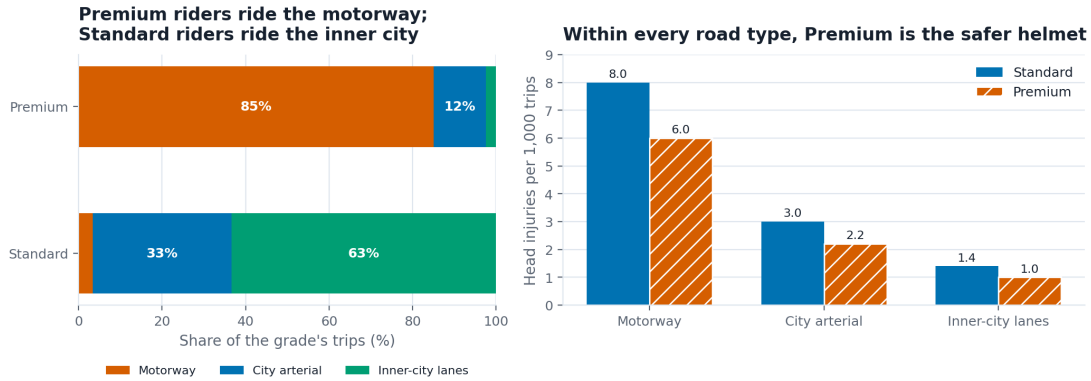
Conclusion: Premium helmets are more dangerous (5.40 vs 2.15 injuries per 1,000 trips).

Decoy: “Premium riders are just riskier: faster or less experienced.” Within a road type the riders are not faster: motorway 72.2 vs 71.0 kph, city arterial 39.6 vs 38.0, inner-city lanes 25.6 vs 24.0. Experience gaps are mixed (motorway Premium 32 vs Standard 41 months), and Premium is still lower in every stratum. The sample is not the issue either: 100,000 trips, 345 injuries.

3 · The mechanism

Road type is a confounder (Simpson's paradox). 85% of Premium trips are on the motorway vs 3.3% of Standard trips (63% of Standard trips are in inner-city lanes). The motorway is intrinsically dangerous (6.1 per 1,000 vs 1.4 in inner-city lanes). Inside each road type Premium is lower: 6.0 vs 8.0, 2.2 vs 3.0, 1.00 vs 1.39. Standardised to the combined road mix: Standard 4.17, Premium 3.10 (-26%).

4 · The redesign



Rates computed as sum(injuries)/sum(trips)*1000 per cell. y from zero. Synthetic data generated for CDB601220 Week 3.

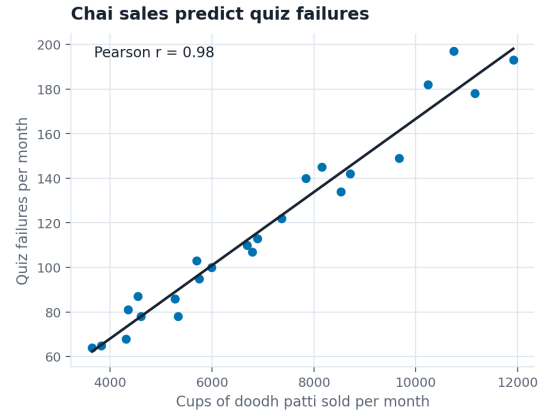
Left: who rides where. Right: rate within each road type. y from zero; rates = sum(injuries)/sum(trips) x 1000.

Case 2 · Chai sales and quiz failures

“Canteen chai sales predict quiz failures.”

1 · Rebuild

Scatter of cups (x) vs failures (y), one point per month, OLS line, r annotated.



One point per month, n = 24, OLS line. Synthetic data generated for CDB601220 Week 3.

2 · The conclusion the chart invites, and the decoy

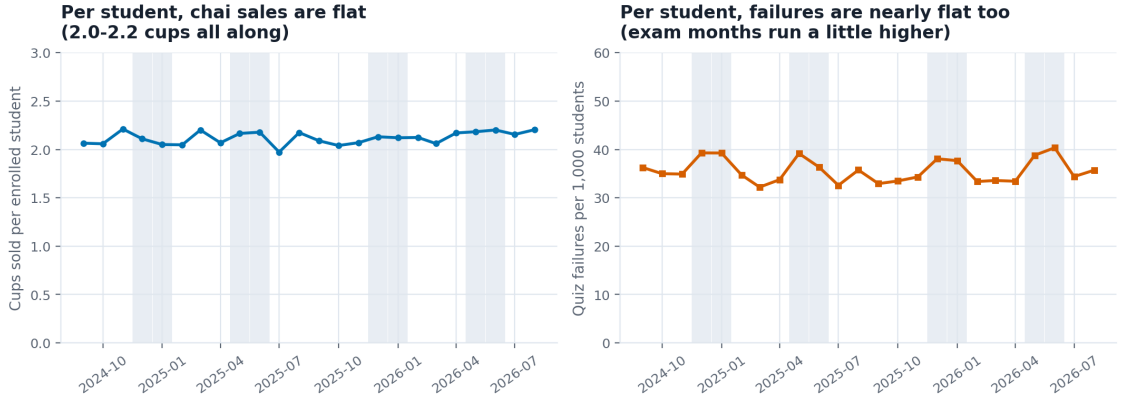
Conclusion: Selling more chai goes with more failed quizzes (r = 0.98), so chai (or whatever drives it) is a lever on failures.

Decoy: “Exam months push up both.” They do not: the correlation of the exam-month flag with cups sold is r = 0.06, and dropping the 8 exam months makes the relationship *stronger* (r = 0.996), not weaker.

3 · The mechanism

Enrolment growth. Students enrolled rose from 1,765 to 5,406 (3.1x). Cups sold track enrolment at r = 0.997 and failures at r = 0.977: both are head-counts that scale with the number of students. Per student the relationship is gone: 1.97-2.21 cups per student in every month, 32-40 failures per 1,000 students, and the correlation between the two rates is only r = 0.29 (p = 0.16, n = 24).

4 · The redesign



Two panels, one measure each, both y-axes from zero. Grey bands = exam months. Rates divided by students_enrolled. Synthetic data generated for CDB601220 Week 3.

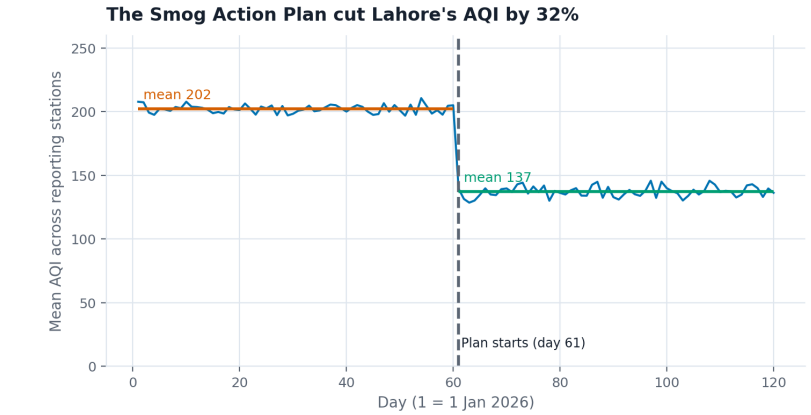
Two panels, one measure each (no dual axis), y from zero. Grey bands mark exam months.

Case 3 · The Smog Action Plan

“The Smog Action Plan cut Lahore's AQI by a third.”

1 · Rebuild

Mean aqi per date over all rows, y from zero, marker at day 61, period mean lines.



Mean aqi per date over all rows (blanks skipped); y from zero. Synthetic data generated for CDB601220 Week 3.

2 · The conclusion the chart invites, and the decoy

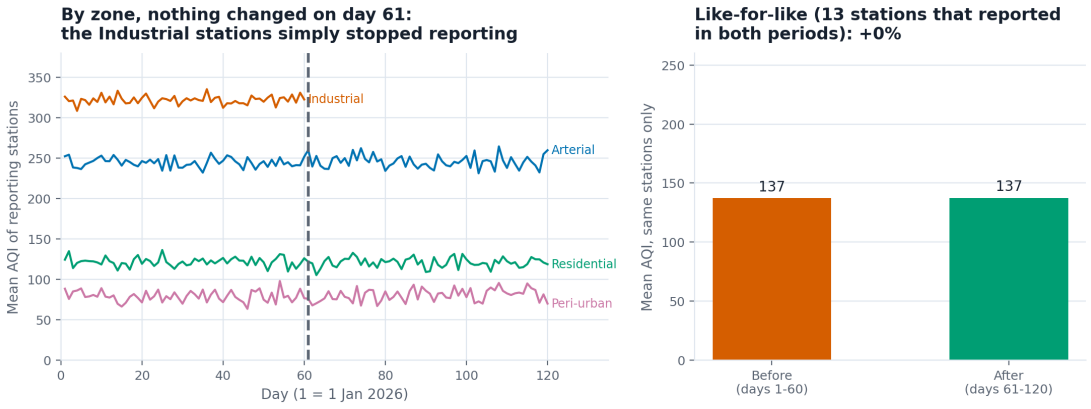
Conclusion: AQI fell from 202 to 137 (-32%) after day 61, so the plan worked.

Decoy: “It is just the season / the weather.” Weather would move every zone. Zones that reported on all 120 days did not move: Residential 121.6 → 120.8, Peri-urban 78.9 → 80.4, Arterial 244.7 → 245.9. (The axis is not truncated either: I set it from zero.)

3 · The mechanism

A change in who is being averaged. 540 cells are blank and pandas skips them silently. From day 61 the 6 Industrial stations (mean AQI 322, the dirtiest zone) are 100% offline and Arterial stations are 50% offline, so the mean is taken over fewer, cleaner stations (average reporting stations per day: 22.0 → 13.0). Like-for-like, over the 13 stations that reported in both periods, AQI is 137.1 → 137.2 (+0.1%): no change at all. The ‘32% fall’ is arithmetic produced by missing data, not by the plan.

4 · The redesign



Lines are broken where a zone has no reporting station. Direct labels, no legend. Like-for-like = stations with data in both periods. Synthetic data generated for CDB601220 Week 3.

Left: one line per zone, direct labels, broken where a zone has no data. Right: same-station comparison.

Case 4 · Four departments, identical statistics

“Four departments behave identically, so fund them equally.”

1 · Rebuild

Not a chart: a table of n, mean, SD, Pearson r and slope per department (sample SD). Also saved to case4_table.csv.

Four departments, identical summary statistics (the analyst's table)

	n	mean x	mean y	SD x	SD y	Pearson r	slope
Computer Science	11	18.000	14.501	6.633	2.032	0.816	0.250
Business Administration	11	18.000	14.501	6.633	2.032	0.816	0.250
Electrical Engineering	11	18.000	14.500	6.633	2.030	0.816	0.250
Architecture	11	18.000	14.501	6.633	2.031	0.817	0.250

Sample SD (ddof = 1). Synthetic data generated for CDB601220 Week 3.

2 · The conclusion the chart invites, and the decoy

Conclusion: Same mean hours (18.00), same mean score (14.50), same r (0.816), same slope (0.25): the departments are interchangeable.

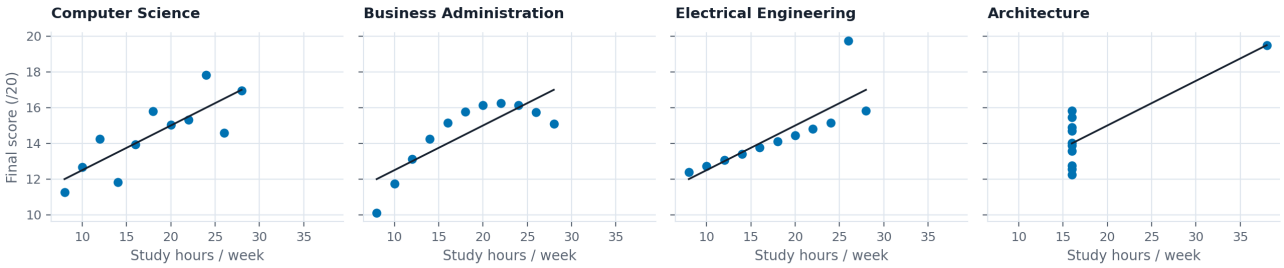
Decoy: “Eleven students per department is too small to say anything.” Sample size is not what breaks this: eleven points are plenty to *see* the difference, and a bigger sample from the same processes would give the same four shapes.

3 · The mechanism

Summary statistics do not constrain shape (Anscombe, 1973). Business Administration: a perfect curve — a quadratic fits with $R^2 = 1.00$ against 0.67 for the line. Electrical Engineering: a straight line broken by one outlier (largest residual 3.24 marks vs about 1.9 elsewhere; Cook's D 1.39). Architecture: ten of eleven students study exactly 16 hours and one studies 38 — only 2 distinct x values, so the whole slope hangs on a single student (leverage 1.0). Computer Science is the only department where the straight-line summary is a fair description.

4 · The redesign

Same line, four different stories: a curve, an outlier, one lever point, and a clean fit



OLS line in each panel; shared axes. Synthetic data generated for CDB601220 Week 3.

One scatter per department with its OLS line, shared axes.

Case 5 · Five cohorts, one set of statistics

“Five cohorts, one set of statistics.”

1 · Rebuild

Not a chart: a table of n, means, SDs (population), r and slope per cohort. Also saved to case5_table.csv.

Five cohorts, one set of statistics (the analyst's table)

	n	mean x	mean y	SD x	SD y	Pearson r	slope
straight line	180	48.000	63.000	12.000	15.000	0.420	0.525
ring	180	48.000	63.000	12.000	15.000	0.420	0.525
star	180	48.000	63.000	12.000	15.000	0.420	0.525
two clusters	180	48.000	63.000	12.000	15.000	0.420	0.525
the letter X	180	48.000	63.000	12.000	15.000	0.420	0.525

Population SD (ddof = 0). Synthetic data generated for CDB601220 Week 3.

2 · The conclusion the chart invites, and the decoy

Conclusion: The five learning-platform cohorts are statistically the same group (mean 48/63, SD 12/15, r = 0.420).

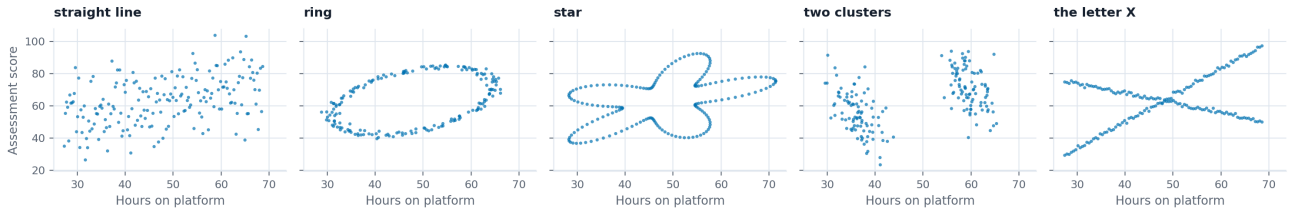
Decoy: “The differences are just sampling noise.” There is no noise to blame: n = 180 per cohort and the statistics agree to within 4e-05, because they were forced to — not estimated.

3 · The mechanism

Five numbers describe a point cloud's centre, spread and linear tilt, not its shape. *vizlib.match_stats()* rescales each cloud linearly to the target mean, SD and r, as in the Datasaurus Dozen (Matejka & Fitzmaurice, 2017). Every cohort therefore has $R^2 = 0.176$ and the same residual SD about the line (13.7), yet the cohorts are a line, a ring, a star, two clusters and the letter X; the excess kurtosis of hours on platform alone runs from -1.78 to -0.94.

4 · The redesign

Same mean, SD and r = 0.420 in all five: a line, a ring, a star, two clusters and the letter X



n = 180 per cohort, shared axes. Synthetic data generated for CDB601220 Week 3.

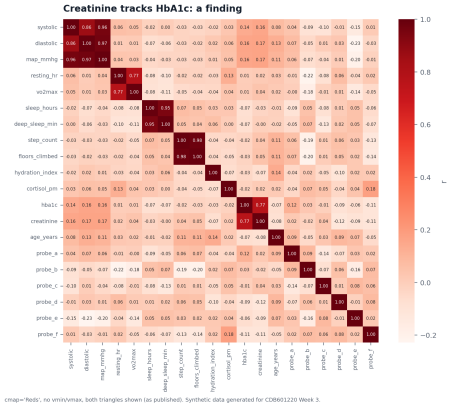
Small multiples, shared axes, n = 180 each.

Case 6 · Creatinine tracks HbA1c

“Creatinine tracks HbA1c — a finding.”

1 · Rebuild

Annotated heatmap of the 20 x 20 correlation matrix, cmap=‘Reds’, no vmin/vmax, both triangles.



2 · The conclusion the chart invites, and the decoy

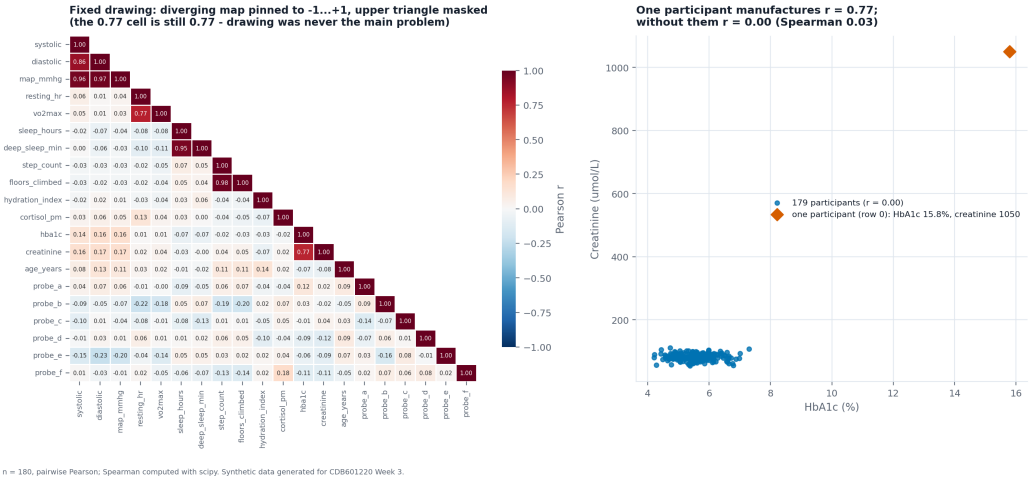
Conclusion: Creatinine and HbA1c are strongly related (r = 0.77), a candidate kidney–glucose link worth reporting.

Decoy: “Both rise with age (a confounder).” Not here: age correlates with HbA1c at r = -0.07 and with creatinine at r = -0.08. And “190 pairs, so some will be flukes” is also not the story: apart from the arithmetic pairs there is no other strong cell.

3 · The mechanism

Two problems, only one of them is the drawing. Drawing: a one-hue sequential map (Reds) for a signed quantity, with no vmin/vmax (the scale stretches from -0.23 to 1) and both triangles shown, so negative and zero look alike and every cell is printed twice. Fixing it (diverging RdBu_r pinned to −1…+1, upper triangle masked) does **not** change the 0.77 cell — it is still 0.77. Finding: one participant (row 0: HbA1c 15.8%, creatinine 1050 µmol/L, against medians of 5.5 and 82) produces the entire correlation. Without that single row r = 0.00; Spearman r = 0.03. Pearson is not robust to one extreme point, and a heatmap hides the scatter that would have shown it. (Also: systolic–MAP, r = 0.96, is arithmetic, and the probes are uncalibrated channels.)

4 · The redesign



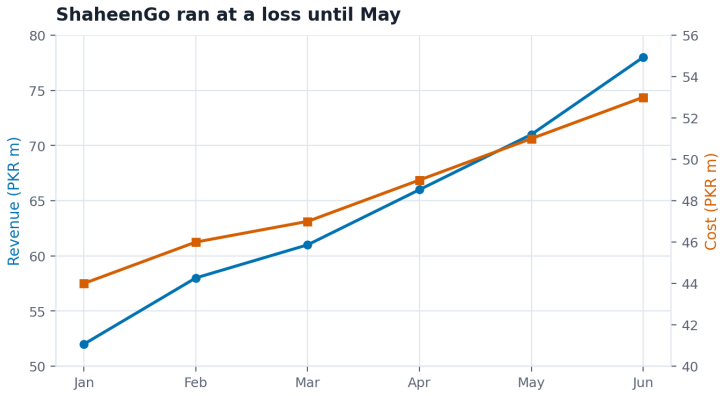
Left: fixed heatmap. Right: the scatter the heatmap hid, with the one participant marked.

Case 7 · ShaheenGo's loss until May

“ShaheenGo ran at a loss until May.”

1 · Rebuild

Line chart with ax.twinx(): revenue on 50–80 (left), cost on 40–56 (right).



Two y-axes via ax.twinx(): 50-80 and 40-56 (as published). Synthetic data generated for CDB601220 Week 3.

2 · The conclusion the chart invites, and the decoy

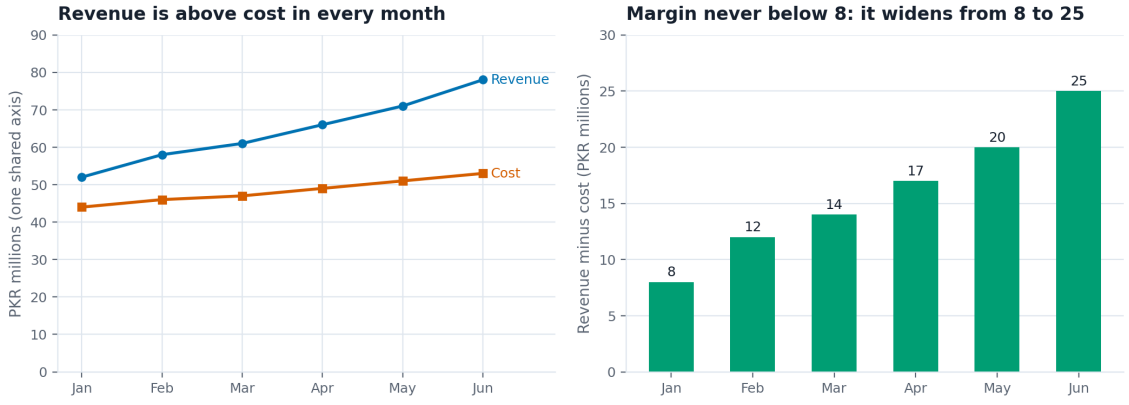
Conclusion: Cost exceeded revenue for the first four months, then revenue overtook it in May.

Decoy: “The numbers must contain a crossover, maybe a one-off cost.” They contain none: both columns are PKR millions and revenue exceeds cost in all six months (margin 8 → 25 PKR m).

3 · The mechanism

Two independent y-axes. Revenue is plotted on 50–80 and cost on 40–56, so each line is stretched to fill the same height; where the lines cross on screen is decided entirely by the windows chosen, not by the data. The cost line is drawn above revenue in 4 of 6 months although revenue was higher in all 6. Revenue grew +50% and cost +20%: the business grew its margin every month.

4 · The redesign



Two panels, no twin axes, both y-axes from zero. Synthetic data generated for CDB601220 Week 3.

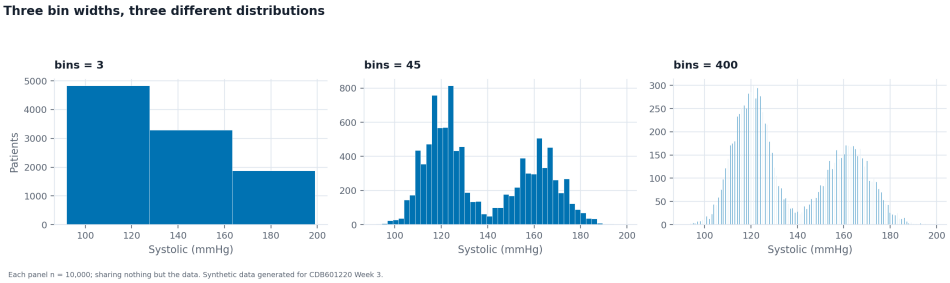
Left: both measures on ONE axis from zero. Right: the margin on its own panel.

Case 8 · Three bin widths

“Three bin widths, three different distributions.”

1 · Rebuild

Three histograms of systolic with bins = 3, 45, 400 (n = 10,000 in each).



2 · The conclusion the chart invites, and the decoy

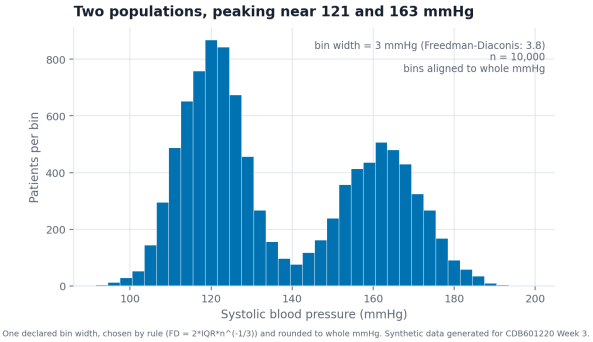
Conclusion: The shape of the systolic distribution depends on who draws it: one hump, a bimodal curve, or a spiky comb.

Decoy: “Different samples, or too few observations.” All three panels use the same 10,000 values (n = 10,000 in each), so it is not sampling.

3 · The mechanism

Bin width is a parameter, and two of these choices are bad. bins = 3 means 36 mmHg per bin, which merges two populations into one hump. bins = 400 means 0.27 mmHg per bin, narrower than the data's 1-mmHg resolution (104 distinct values), so 74% of the bins are empty and the histogram is a comb. bins = 45 (2.4 mmHg) happens to work. A rule beats luck: Freedman–Diaconis gives 3.8 mmHg; I use 3 mmHg, aligned to whole mmHg.

4 · The redesign



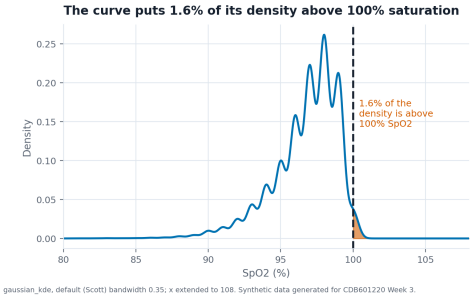
One histogram with a declared bin width, aligned to whole mmHg; peaks near 121 and 163.

Case 9 · Density above 100% saturation

“4.4% of the density sits above 100% saturation.”

1 · Rebuild

KDE of spo2, default bandwidth, x extended to 108, line at 100.



2 · The conclusion the chart invites, and the decoy

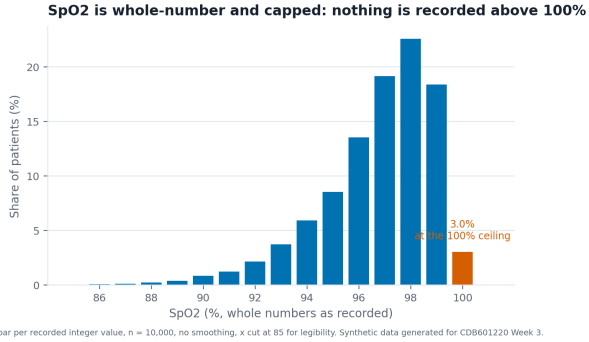
Conclusion: Some patients have an oxygen saturation above 100%, so the data (or the instruments) are broken.

Decoy: “The data contain impossible readings.” They do not: the maximum is 100; 304 patients (3.0%) sit exactly at 100.

3 · The mechanism

The estimator, not the data, leaks past a hard ceiling. A Gaussian KDE puts a symmetric kernel (default bandwidth 0.35 points) on every observation; the 304 readings at the ceiling each spill half their mass above 100. I measure 1.6% of the curve's area above 100 (4.3% above 99.5); the brief's 4.4% did not reproduce exactly with the default bandwidth, so I report what I measured. SpO2 is also whole-number, so a smooth curve invents detail that was never recorded.

4 · The redesign



One bar per recorded whole-number value; nothing exists above 100.

References: Anscombe (1973) The American Statistician 27(1); Matejka & Fitzmaurice (2017) Same Stats, Different Graphs; Tufte (1983) The Visual Display of Quantitative Information; Wilke, Fundamentals of Data Visualization; seaborn and SciPy documentation. All data synthetic, CDB601220 Week 3.